RAJALAKSHMI ENGINEERING COLLEGE

Ragaventhara Raja R K 251801253-2025-AIDS

R2

Started on	Tuesday, 25 August 2026, 4:35 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 12:03 PM
Time taken	19 hours 27 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    int count = 0;

    if (scanf("%d", &n) != 1) return 0;

    // 1. Step for: int rev = 0;
    count++;

    // 2. While loop structure
    while (n != 0) {
        count++; // For while (n != 0) condition check

        count++; // For remainder = n % 10;
        count++; // For rev = rev * 10 + remainder;
        count++; // For n /= 10;

        // Simulating the loop body
        n /= 10;
    }

    // 3. Step for final false condition check of while (n != 0)
    count++;

    // 4. Step for print(rev);
    count++;

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

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R2

Started on	Tuesday, 25 August 2026, 3:53 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 12:04 PM
Time taken	20 hours 11 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    int count = 0;

    if (scanf("%d", &n) != 1) return 0;

    // 1. Step count for initialization: c = 0
    count++;

    // 2. Step counting across nested loops
    for (int i = n / 2; i < n; i++) {
        count++; // Outer loop condition check (i < n)

        for (int j = 1; j < n; j = 2 * j) {
            count++; // Middle loop condition check (j < n)

            for (int k = 1; k < n; k = k * 2) {
                count++; // Inner loop condition check (k < n)
                count++; // Statement execution (c++)
            }
            count++; // Inner loop final false check
        }
        count++; // Middle loop final false check
    }
    count++; // Outer loop final false check

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

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Ragaventhara Raja R K 251801253-2025-AIDS

R2

Started on	Tuesday, 25 August 2026, 3:45 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:48 AM
Time taken	20 hours 3 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void Factor(int num, int *count) {
    int i;
    for (i = 1; i <= num; ++i) {
        (*count) += 2;

        if (num % i == 0) {
            (*count)++;
        }
    }
    (*count)++;
}

int main() {
    int num;
    int count = 0;

    if (scanf("%d", &num) == 1) {
        Factor(num, &count);
        printf("%d", count);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Ragaventhara Raja R K 251801253-2025-AIDS

R2

Started on	Tuesday, 25 August 2026, 3:39 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:49 AM
Time taken	20 hours 9 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void func(int n, int *count) {
    (*count)++;

    if (n == 1) {
        (*count)++;
    } else {

        for (int i = 1; (*count)++, i <= n; i++, (*count)++) {
            for (int j = 1; (*count)++, j <= n;) {
                (*count)++;
                (*count)++;
                break;
            }
        }
    }
}

int main() {
    int n;
    int count = 0;
    if (scanf("%d", &n) == 1) {
        func(n, &count);
        printf("%d", count);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

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Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int count = 0;

    int i = 1; count++;
    int s = 1; count++;

    while (count++, s <= n) {
        i++; count++;
        s += i; count++;
    }

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course